LAB MANUAL

OBJECT ORIENTED PROGRAMMING
23CSE111

AMRITA SCHOOL OF COMPUTING

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	• Create another constructor that accepts memberId, memberName, membershipType. • Display the member details.	
02.	An e-commerce system maintains customer carts. Class Name: Cart Attributes: • String customerName • int productCount • double totalPrice Task: • Create a constructor that sets default values customerName = "Guest", productCount = 0, totalPrice = 0. • Create another constructor that accepts all three values. • Write a method to display cart details.	
03.	A smart device manufacturing company initializes devices. Class Name: Device Attributes: • String deviceName • String status • String firmwareVersion Task: • Create a default constructor with deviceName = "SmartDevice", status = "Inactive", firmwareVersion = "v1.0". • Create another constructor that accepts deviceName and firmwareVersion. • Display • device information.	
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EXPERIMENT-1
BASIC JAVA PROGRAMS

01. Write a Java Program to Print Hello and your Name in separate lines.

CODE:

```
import java.util.Scanner;
public class J1
{
  public static void main(String[] args)
{
  Scanner sc=new Scanner(System.in);
  String name=sc.nextLine();
  System.out.println("Welcome"+name);
  sc.close();
}
}
```

OUTPUT:

```
D:\OOPS\BASIC JAVA PROGRAMS>javac J1.java

D:\OOPS\BASIC JAVA PROGRAMS>java J1
Nisha
Welcome Nisha
```

02. Write a java program to print sum, difference, product and division of two numbers.

CODE:

```
import java.util.Scanner;
class J2
{
  public static void main(String[] args)
{
  Scanner sc = new Scanner(System.in);
  int a = sc.nextInt();
  int b = sc.nextInt();
  System.out.println("Sum = " + (a + b));
  System.out.println("Difference = " + (a - b));
  System.out.println("Product = " + (a * b));
  System.out.println("Division = " + (a / b));
}
}
```

OUTPUT:

```
D:\OOPS\BASIC JAVA PROGRAMS>javac J2.java

D:\OOPS\BASIC JAVA PROGRAMS>java J2
25
5
Sum = 30
Difference = 20
Product = 125
Division = 5
```

03. Write a java program to find the number is odd or even number.

CODE:

```
import java.util.Scanner;
public class J3
{
public static void main(String[] args)
{
Scanner sc=new Scanner(System.in);
int n=sc.nextInt();
if (n%2==0)
System.out.println(n + " is a Even number");
else
System.out.println(n + "is a odd number");
}
}
```

OUTPUT:

```
D:\OOPS\BASIC JAVA PROGRAMS>javac J3.java

D:\OOPS\BASIC JAVA PROGRAMS>java J3
23
23is a odd number

D:\OOPS\BASIC JAVA PROGRAMS>javac J3.java

D:\OOPS\BASIC JAVA PROGRAMS>java J3
18
18 is a Even number
```

04. Write a java program to find largest of two numbers.

CODE:

```
import java.util.Scanner;
public class J4
{
public static void main(String[] args)
{
Scanner sc=new Scanner(System.in);
System.out.print("Enter the first number :");
int a=sc.nextInt();
System.out.print("Enter the second number : ");
int b=sc.nextInt();
if (a>b)
System.out.println(a + " is the largest number");
else
System.out.println(b + " is the largest number");
}
}
```

OUTPUT:

```
D:\OOPS\BASIC JAVA PROGRAMS>javac J4.java

D:\OOPS\BASIC JAVA PROGRAMS>java J4
Enter the first number :1000
Enter the second number : 2000
2000 is the largest number
```

05. Write a java program to check the number is positive, negative or zero.

CODE:

```
import java.util.Scanner;
public class J5{
public static void main(String[] args){
Scanner sc=new Scanner(System.in);
int n=sc.nextInt();
if (n>0)
System.out.println(n + "is a positive number");
else if (n<0)
System.out.println(n + "is a negative number");
else
System.out.println(n + "is Zero");
}
}
```

OUTPUT:

```
D:\OOPS\BASIC JAVA PROGRAMS>javac J5.java

D:\OOPS\BASIC JAVA PROGRAMS>java J5
100
100is a positive number

D:\OOPS\BASIC JAVA PROGRAMS>javac J5.java

D:\OOPS\BASIC JAVA PROGRAMS>java J5
-100
-100is a negative number

D:\OOPS\BASIC JAVA PROGRAMS>javac J5.java

D:\OOPS\BASIC JAVA PROGRAMS>java J5
0
0is Zero
```

06. Write a java program to print numbers from 1 to N using for loop.

CODE:

```
import java.util.Scanner;
public class J6
{
public static void main(String[] args)
{
Scanner sc=new Scanner(System.in);
int n=sc.nextInt();
for (int i=1;i<=n;i++)
System.out.println(i);
}
}
```

OUTPUT:

```
D:\OOPS\BASIC JAVA PROGRAMS>javac J6.java

D:\OOPS\BASIC JAVA PROGRAMS>java J6
10
1
2
3
4
5
6
7
8
9
10
```

07. Write a java program to print the sum of first N Natural numbers.

CODE:

```
import java.util.Scanner;
public class J7
{
public static void main(String[] args)
{
Scanner sc=new Scanner(System.in);
int n=sc.nextInt();
int sum=0;
for (int i = 1; i <= n; i++)
sum += i;
System.out.println("Sum : "+sum);
}
}
```

OUTPUT:

```
D:\OOPS\BASIC JAVA PROGRAMS>javac J7.java

D:\OOPS\BASIC JAVA PROGRAMS>java J7
10
Sum : 55
```

08. Write a java program to print the factorial of the number.

CODE:

```
import java.util.Scanner;
public class J8
{
public static void main(String[] args)
{
Scanner sc=new Scanner(System.in);
int n=sc.nextInt();
int fact=1;
for (int i=1;i<=n;i++)
    fact*=i;
System.out.println("Factorial = "+fact);
}
}
```

OUTPUT:

```
D:\OOPS\BASIC JAVA PROGRAMS>javac J8.java

D:\OOPS\BASIC JAVA PROGRAMS>java J8
10
Factorial = 3628800
```

09. Write a java program to reverse a number using while loop.

CODE:

```
import java.util.Scanner;
public class J9{
public static void main(String[] args)
{
Scanner sc=new Scanner(System.in);
System.out.print("Enter a number : ");
int n =sc.nextInt();
int rev=0;
while(n!=0)
{
int digit=n%10;
rev=rev*10+digit;
n=n/10;
}
System.out.println("Reversed = "+rev);
}
}
```

OUTPUT:

```
D:\OOPS\BASIC JAVA PROGRAMS>java J9
Enter a number : 15
Reversed = 51
```

10. Write a java program to count the digits of a number

CODE:

```
import java.util.Scanner;
public class J10 {
public static void main(String[] args) {
Scanner sc = new Scanner(System.in);
System.out.print("Enter a number:");
int n = sc.nextInt();
int count=0;
while (n != 0) {
n = n / 10;
count++;
}
System.out.println("Digits = " + count);
}
}
```

OUTPUT:

```
D:\OOPS\BASIC JAVA PROGRAMS>java J10
Enter a number:23876
Digits = 5
```

11. Write a java program to check whether the number is prime number.

CODE:

```
import java.util.Scanner;
public class J11{
public static void main(String[] args){
Scanner sc=new Scanner(System.in);
System.out.print("Enter a number : ");
int n=sc.nextInt();
boolean prime=true;
if (n<=1)prime=false;
for(int i=2;i<=n/2;i++)
if (n%i==0)
prime=false;
break;
if(prime)
System.out.println(n+" is a prime number");
else
System.out.print(n+" is not a prime number");
}
}
```

OUTPUT:

```
D:\OOPS\BASIC JAVA PROGRAMS>javac J11.java

D:\OOPS\BASIC JAVA PROGRAMS>java J11
Enter a number : 3
3 is a prime number

D:\OOPS\BASIC JAVA PROGRAMS>javac J11.java

D:\OOPS\BASIC JAVA PROGRAMS>java J11
Enter a number : 10
10 is not a prime number
```

12. Write a java program to print the Fibonacci series using for loop.

CODE:

```
import java.util.Scanner;
public class J12
{
public static void main(String[] args)
{
Scanner sc=new Scanner(System.in);
int n=sc.nextInt();
int a=0,b=1;
for (int i=1;i<=n;i++)
{
System.out.println(a);
int c=a+b;
a=b;
b=c;
}
}
}
```

OUTPUT:

```
D:\OOPS\BASIC JAVA PROGRAMS>javac J12.java

D:\OOPS\BASIC JAVA PROGRAMS>java J12
0
1
1
2
3
5
8
13
21
34
```

13. Write a java program to print multiplication table using for loop.

CODE:

```
import java.util.Scanner;
public class J13{
public static void main(String[] args)
{
Scanner sc=new Scanner(System.in);
System.out.print("Table-");
int n=sc.nextInt();
for (int i=1;i<=10;i++)
System.out.println(n+"x"+i+"="+(n*i));
}
}
```

OUTPUT:

```
D:\OOPS\BASIC JAVA PROGRAMS>javac J13.java

D:\OOPS\BASIC JAVA PROGRAMS>java J13
Table-4
4x1=4
4x2=8
4x3=12
4x4=16
4x5=20
4x6=24
4x7=28
4x8=32
4x9=36
4x10=40
```

14. Write a java program to calculate the grade using conditional statements.

CODE:

```
import java.util.Scanner;
public class J14
{
public static void main(String[] args)
{
Scanner sc=new Scanner(System.in);
System.out.print("Enter the marks : ");
int marks=sc.nextInt();
if(marks>=90)
System.out.print("A");
else if(marks>=75)
System.out.print("B");
else if(marks>=50)
System.out.print("C");
else
System.out.print("Fail");
}
}
```

OUTPUT:

```
D:\OOPS\BASIC JAVA PROGRAMS>javac J14.java

D:\OOPS\BASIC JAVA PROGRAMS>java J14
Enter the marks : 86
B
```

15. Write a java program to check the login is success or failed

CODE:

```
import java.util.Scanner;
public class J15
{
public static void main(String[] args)
{
Scanner sc=new Scanner(System.in);
System.out.print("Enter username : ");
String user=sc.nextLine();
System.out.print("Enter Password : ");
String password=sc.nextLine();

if(user.equals("admin")&&password.equals("admin@123"))
System.out.println("Login Success");
else
System.out.println("Login Failed");
}
}
```

OUTPUT:

```
D:\OOPS\BASIC JAVA PROGRAMS>javac J15.java

D:\OOPS\BASIC JAVA PROGRAMS>java J15
Enter username : admin
Enter Password : admin@123
Login Success
```

16. Write a java program to check the leap year using conditional statements.

CODE:

```
import java.util.Scanner;
public class J16
{
public static void main(String[] args)
{
Scanner sc=new Scanner(System.in);
int year=sc.nextInt();
if((y%4==0 && y%100 !=0)||y%400==0)
System.out.println("Leap Year");
else
System.out.print("Not Leap Year");
}
}
```

OUTPUT:

```
D:\OOPS\BASIC JAVA PROGRAMS>javac J16.java

D:\OOPS\BASIC JAVA PROGRAMS>java J16
Enter the year : 2008
Leap Year

D:\OOPS\BASIC JAVA PROGRAMS>javac J16.java

D:\OOPS\BASIC JAVA PROGRAMS>java J16
Enter the year : 2006
Not Leap Year
```

17. Write a java program to sum the numbers until the user enters 0 (while loop).

CODE:

```
import java.util.Scanner;
public class J17 {
public static void main(String[] args) {
Scanner sc = new Scanner(System.in);
int number;
int sum = 0;
System.out.println("Enter numbers (Enter 0 to stop : ")
number = sc.nextInt();
while (number != 0)
{
 sum = sum + number;
 number = sc.nextInt();
 }
 System.out.println("Sum of entered numbers: " + sum);
 sc.close();
}
}
```

OUTPUT:

```
D:\OOPS\Basic Program>java J17.java
Enter numbers (Enter 0 to stop :
1
2
3
4
0
Sum of entered numbers: 10
```

18. Write a java program using switch case for simple calculator.

CODE:

```
import java.util.Scanner;
public class J18 {
public static void main(String[] args) {
Scanner sc = new Scanner(System.in);
System.out.print("Enter first number: ");
double num1 = sc.nextDouble();
System.out.print("Enter second number: ");
double num2 = sc.nextDouble();
System.out.println("Choose operation (+, -, *, /): ");
char operator = sc.next().charAt(0);
double result;
switch (operator) {
case '+':
result = num1 + num2;
System.out.println("Result: " + result);
break;
case '-':
result = num1 - num2;
System.out.println("Result: " + result);
break;
case '*':
result = num1 * num2;
System.out.println("Result: " + result);
break;
case '/':
if (num2 != 0)
System.out.println("Result: " + (num1 / num2));
else
System.out.println("Division by zero is not allowed");
break;
default:
System.out.println("Invalid operator");
}
sc.close();
}
}
```

OUTPUT:

```
D:\OOPS\Basic Program>javac J18.java

D:\OOPS\Basic Program>java J18
Enter first number: 20
Enter second number: 10
Choose operation (+, -, *, /):
+
Result: 30.0

D:\OOPS\Basic Program>javac J18.java

D:\OOPS\Basic Program>java J18
Enter first number: 30
Enter second number: 40
Choose operation (+, -, *, /):
*
Result: 1200.0
```

19. Write a java program using switch case for Month of the year.

CODE:

```
import java.util.Scanner;
public class J19 {
public static void main(String[] args) {
Scanner sc = new Scanner(System.in);
System.out.print("Enter month number (1-12): ");
int month = sc.nextInt();
switch(month) {
case 1:
System.out.println("January");
break;
case 2:
System.out.println("February");
break;
case 3:
System.out.println("March");
break;
case 4:
System.out.println("April");
break;
case 5:
System.out.println("May");
break;
case 6:
System.out.println("June");
break;
case 7:
System.out.println("July");
break;
case 8:
System.out.println("August");
break;
case 9:
System.out.println("September");
break;
case 10:
System.out.println("October");
break;
case 11:
System.out.println("November");
```

```
break;
case 12:
System.out.println("December");
break;
default:
System.out.println("Invalid month number");
}
sc.close();
}
}
```

OUTPUT:

```
D:\OOPS\Basic Program>javac J19.java

D:\OOPS\Basic Program>java J19
Enter month number (1-12): 2
February

D:\OOPS\Basic Program>javac J19.java

D:\OOPS\Basic Program>java J19
Enter month number (1-12): 8
August
```

20. Write a java program using switch case for Day of the week.

CODE:

```
import java.util.Scanner;
public class J20{
public static void main(String[] args) {
Scanner sc = new Scanner(System.in);
System.out.print("Enter day number (1-7): ");
int day = sc.nextInt();
switch(day) {
case 1:
System.out.println("Sunday");
break;
case 2:
System.out.println("Monday");
break;
case 3:
System.out.println("Tuesday");
break;
case 4:
System.out.println("Wednesday");
break;
case 5:
System.out.println("Thursday");
break;
case 6:
System.out.println("Friday");
break;
case 7:
System.out.println("Saturday");
break;
default:
System.out.println("Invalid input! Please enter a number between 1 and 7.");
}
sc.close();
}
}
```

OUTPUT:

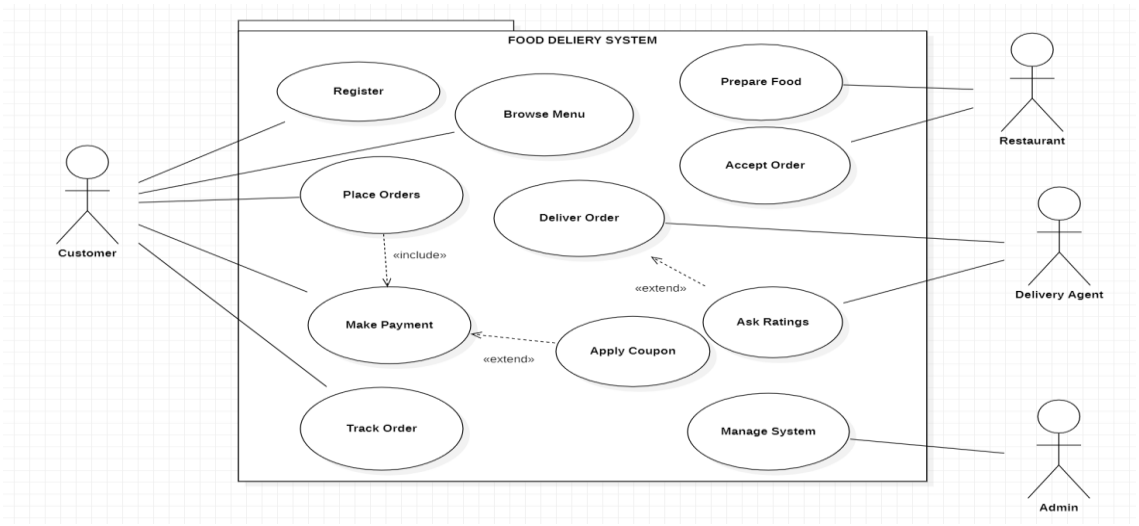
```
D:\OOPS\Basic Program>javac J20.java

D:\OOPS\Basic Program>java J20
Enter day number (1-7): 2
Monday

D:\OOPS\Basic Program>javac J20.java

D:\OOPS\Basic Program>java J20
Enter day number (1-7): 4
Wednesday
```

EXPERIMENT-2
UML DIAGRAMS

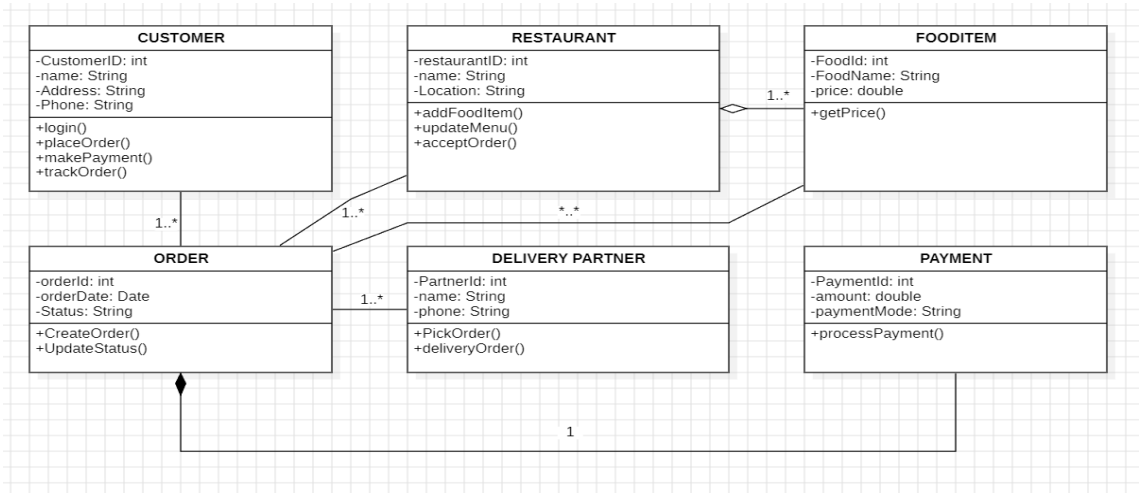
01. To draw UML diagram for Food Delivery system:

- Use Case Diagram
- Class Diagram
- Object Diagram
- Sequence Diagram

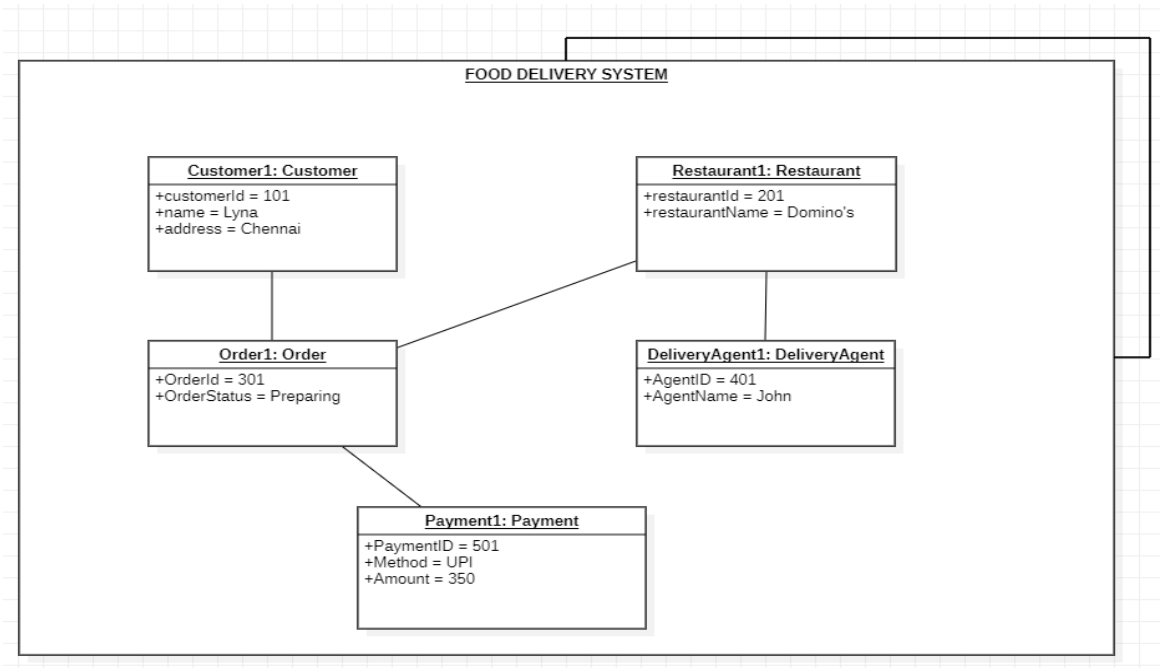
- Use Case Diagram:



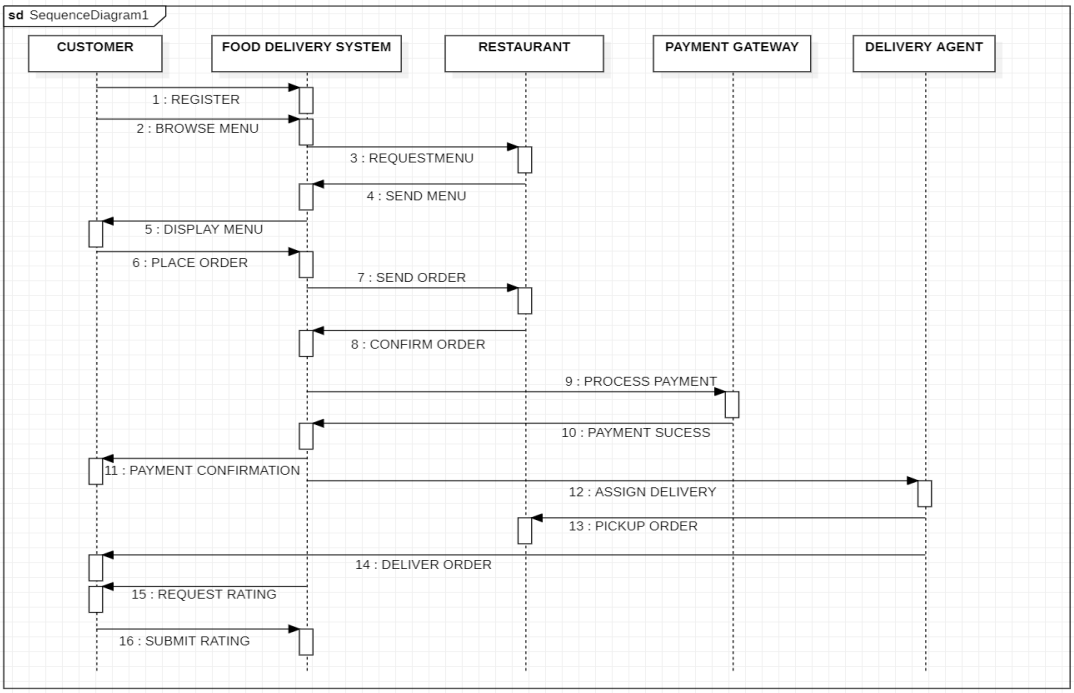
- Class Diagram:



- **Object Diagram:**



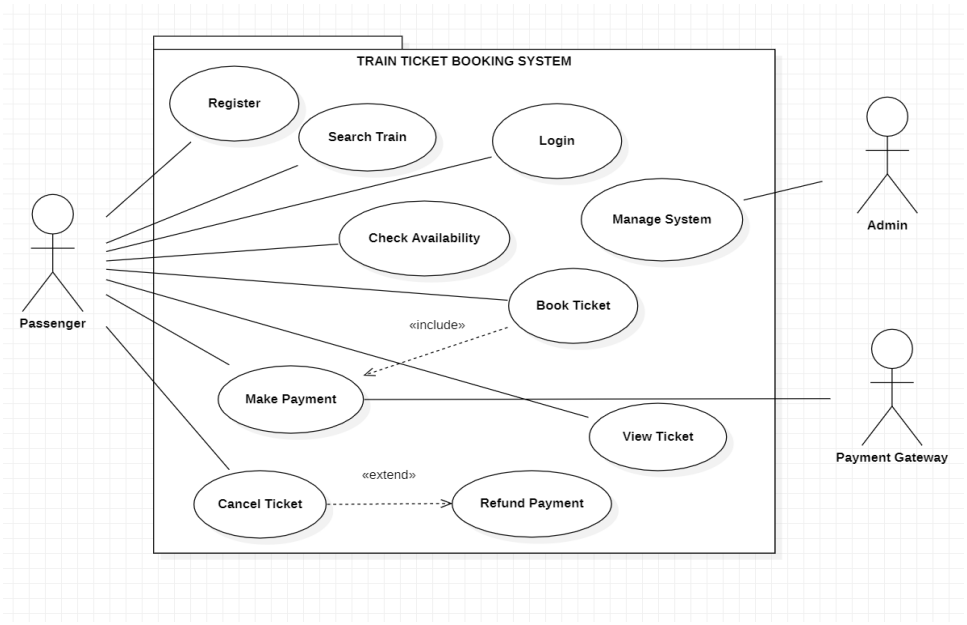
- **Sequence Diagram:**



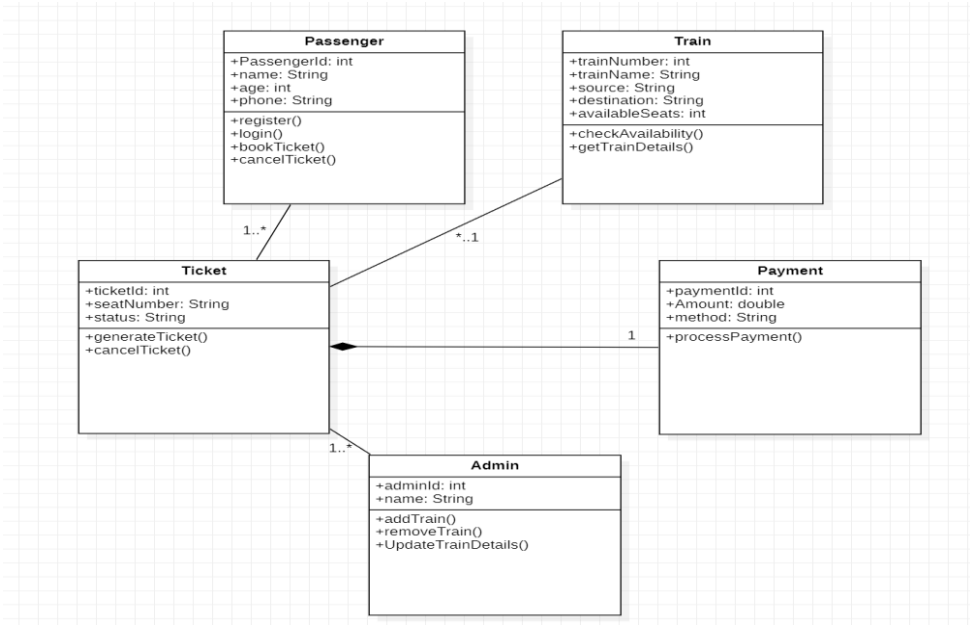
02. To draw UML diagram for Train Ticket booking System:

- Use Case Diagram
- Class Diagram
- Object Diagram
- Sequence Diagram

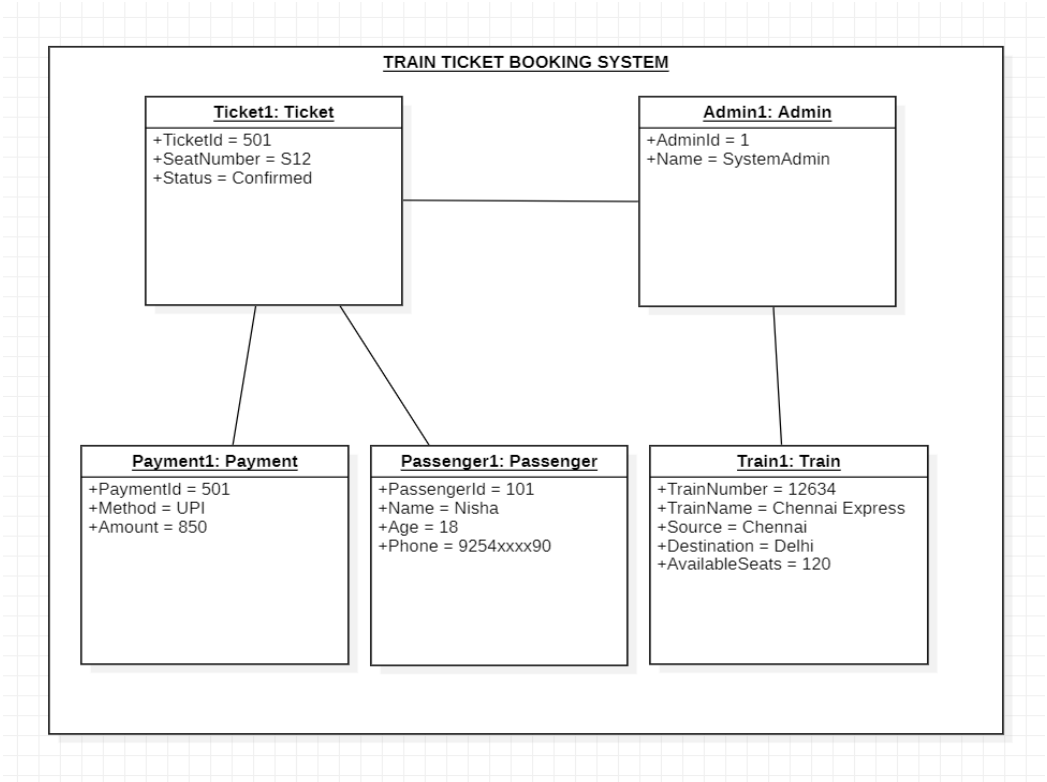
- Use Case Diagram:



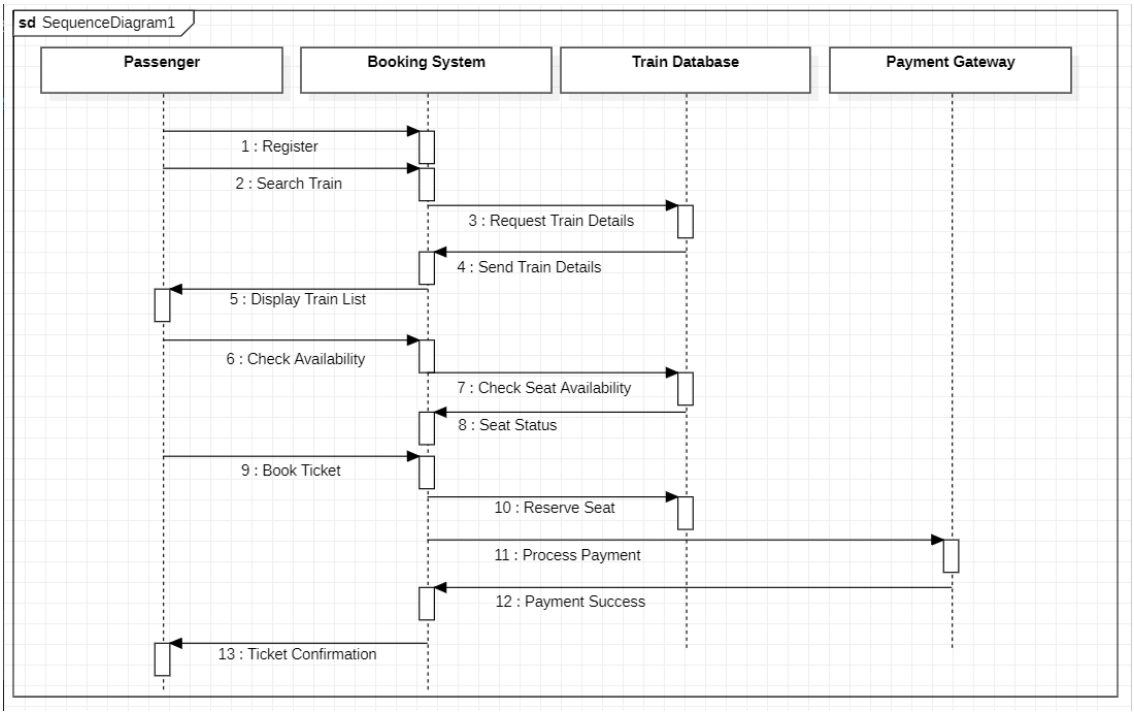
- Class Diagram:



- Object Diagram:



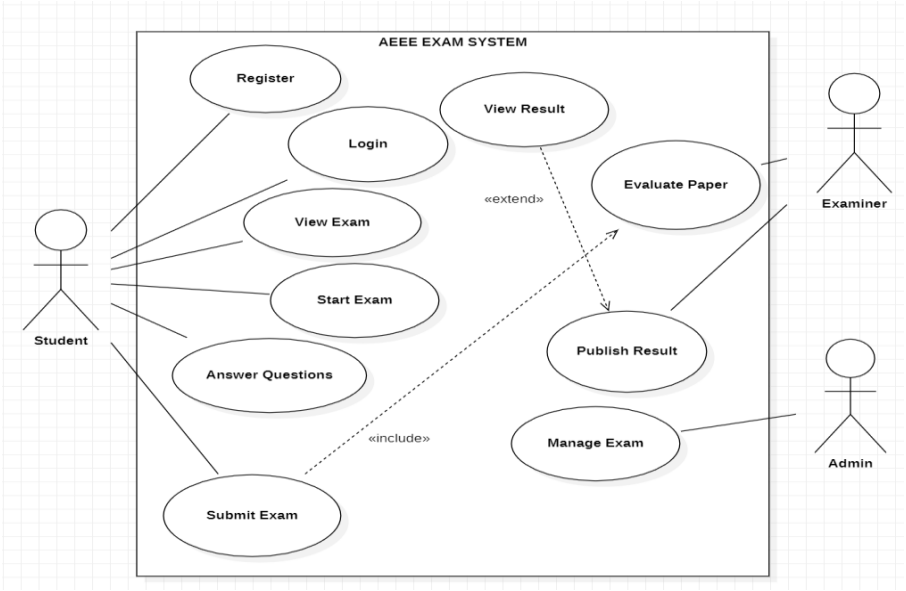
- Sequence Diagram:



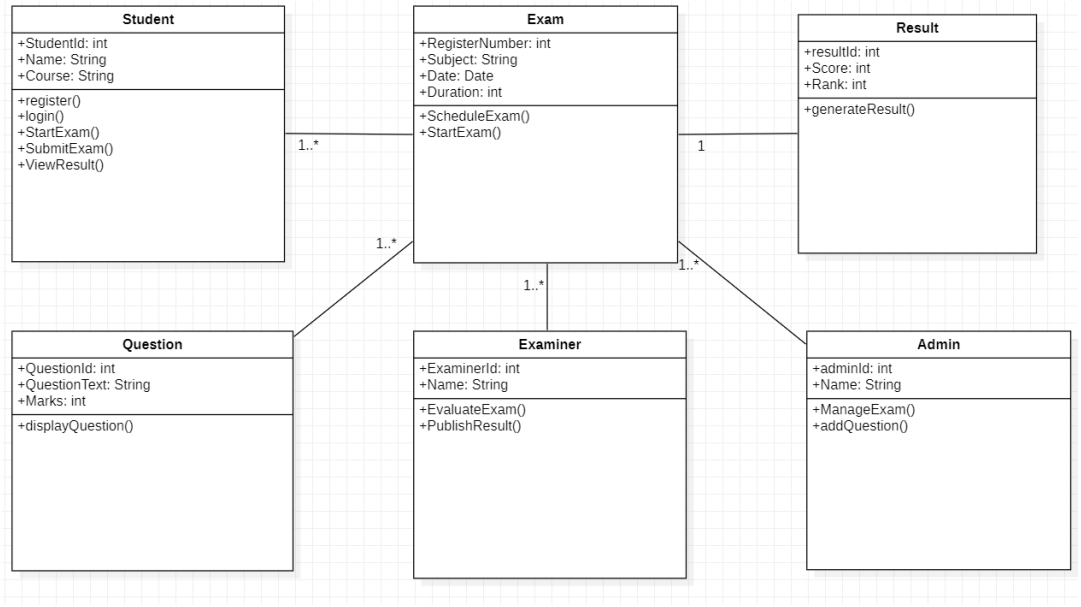
03. To draw UML diagram for AEEE exam system:

- Use Case Diagram
- Class Diagram
- Object Diagram
- Sequence Diagram

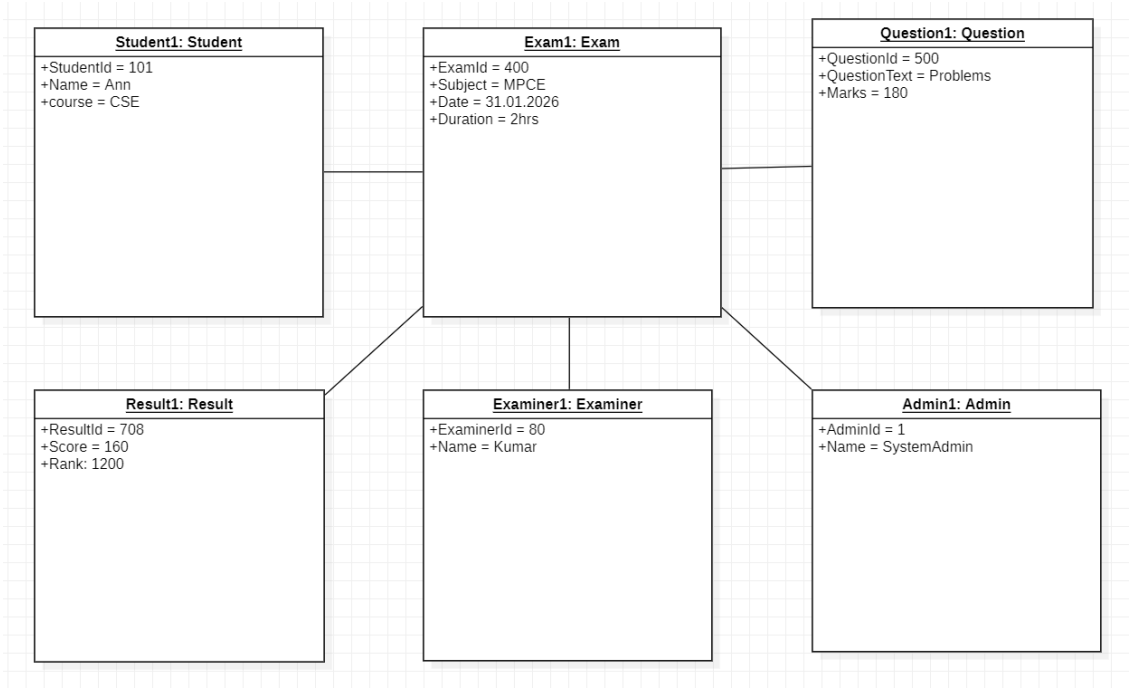
- **Use Case Diagram:**



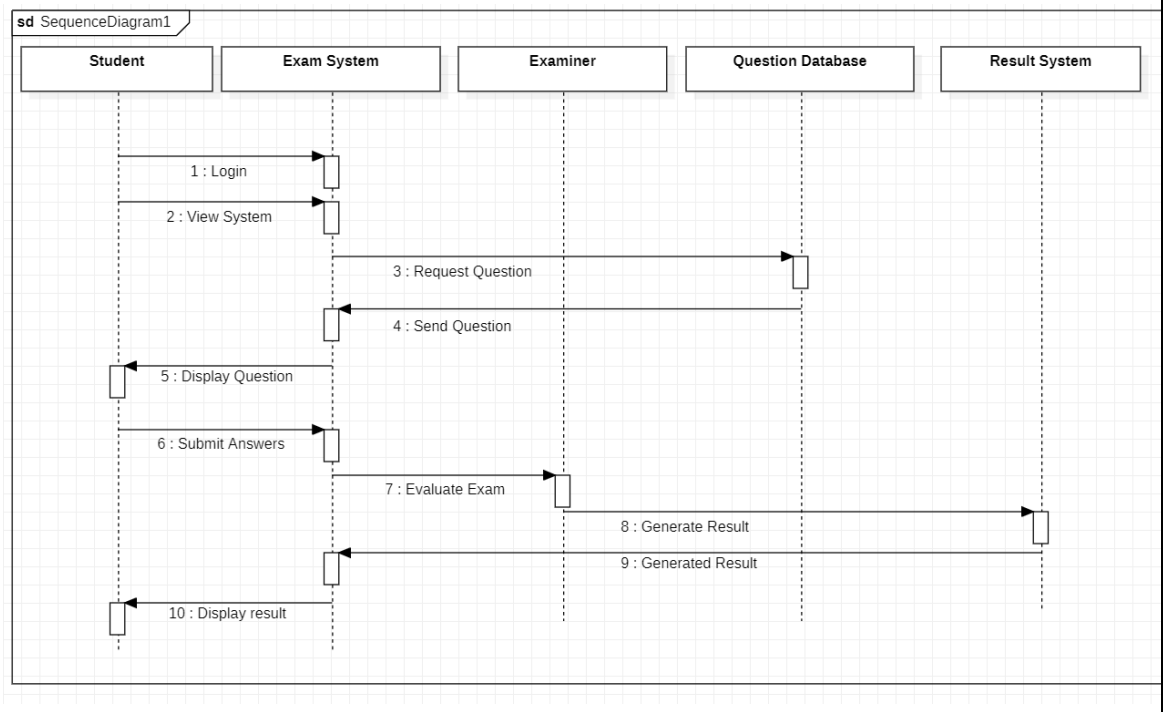
- **Class Diagram:**



- Object Diagram:



- Sequence Diagram:



EXPERIMENT-4
INHERITANCE

01) A company wants to develop a payroll management system to store employee details and calculate the salary of managers.
Since a manager is also an employee, the system should use inheritance to reuse the employee information.
You are asked to implement this system using Java OOPS concepts.
Base Class: Employee
Attributes (Data Members)

- empId – Employee ID
- name – Employee name
- basicSalary – Basic salary of the employee

Method

- displayEmployee()
  → Displays the employee ID, name, and basic salary.

Derived Class: Manager (inherits from Employee)
Additional Attribute

- bonus – Extra bonus given to the manager.

Methods

- calculateSalary()
  → Calculates the total salary using the formula:
  totalSalary = basicSalary + bonus
- displayManager()
  → Displays the manager details including:
  - Employee ID
  - Name
  - Basic Salary
  - Bonus
  - Total Salary

CODE:

```
import java.util.Scanner;
class Employee{
    int empId;
    String Name;
    double basicSalary;
    public Employee(int E1,String E2,double E3){
        this.empId=E1;
        this.Name=E2;
        this.basicSalary=E3;
    }
    void displayEmployee(){
      System.out.println("Employee ID : "+empId);
      System.out.println("Name : "+Name);
      System.out.println("Basic Salary : "+basicSalary);
    }
}
class Manager extends Employee{
    double Bonus;
    public Manager(int E1,String E2,double E3,double E4){
    super(E1,E2,E3);
    this.Bonus=E4;
    }
   void displayManager(){
    super.displayEmployee();
    System.out.println("Bonus Salary : "+Bonus);
   }
   void calculateSalary(double totalSalary){
    super.displayEmployee();
    totalSalary=Bonus+basicSalary;
    System.out.println("Total Salary : "+totalSalary);
   }

}
public class I5 {
    public static void main(String[] args){
    Scanner sc=new Scanner(System.in);
    System.out.print("Enter the Employee ID :");
    int Emp_ID=sc.nextInt();
    sc.nextLine();
    System.out.print("Enter the Name : ");
    String Name=sc.nextLine();
```

```
    System.out.print("Enter the Basic Salary : ");
    double Basic_Salary=sc.nextDouble();
    Employee E=new Employee(Emp_ID,Name,Basic_Salary);
    E.displayEmployee();
    System.out.print("Enter Bonus Salary :" );
    double Bonus=sc.nextDouble();
    Manager M=new Manager(Emp_ID,Name,Basic_Salary,Bonus);
    M.displayManager();

    sc.close();
    }
}
```

OUTPUT:

```
D:\OOPS\INHERITANCE>java I5
Enter the Employee ID :2523
Enter the Name : Ann
Enter the Basic Salary : 200000
Employee ID : 2523
Name : Ann
Basic Salary : 200000.0
Enter Bonus Salary :30000
Employee ID : 2523
Name : Ann
Basic Salary : 200000.0
Bonus Salary : 30000.0
```

02) A vehicle rental company wants to create a Java application to manage different types of vehicles available for rent.
Since all vehicles share some common properties, the company decides to use inheritance to avoid code duplication.
The system should store general vehicle information and also handle specific details for cars and bikes.

Base Class: Vehicle
Create a class named Vehicle with the following attributes:
Attributes
- brand – Brand of the vehicle
- model – Model of the vehicle
- rentalPrice – Rental price per day

Method
- display_Info()
  → Displays the vehicle brand, model, and rental price.

Derived Class: Car (inherits from Vehicle)
Create a class Car that extends the Vehicle class.
Additional Attribute
- fuelType – Type of fuel used (Petrol/Diesel/Electric)

Method
- Override display_Info()
  → Display vehicle details along with the fuel type.

Derived Class: Bike (inherits from Vehicle)
Create another class Bike that extends the Vehicle class.
Additional Attribute
- engineCapacity – Engine capacity of the bike (in cc)

Method

CODE:

```
import java.util.Scanner;
class Vehicle{
String brand;
String model;
int Rental_Price;
public Vehicle(String a,String b,int c){
this.brand=a;
this.model=b;
this.Rental_Price=c;
}
void display_Info(){
System.out.println("BRAND : "+brand);
System.out.println("MODEL : "+model);
System.out.println("RENTAL PRICE : "+Rental_Price);
}}
class Car extends Vehicle{
String Fuel_type;
public Car(String a,String b,int c,String d){
super(a,b,c);
this.Fuel_type=d;
}
void display_Info(){
super.display_Info();
System.out.print("FUEL TYPE :"+Fuel_type);
}}
class Bike extends Vehicle{
int Engine_Capacity;
public Bike(String a,String b,int c,int d){
super(a,b,c);
this.Engine_Capacity=d;
}
void display_Info(){
super.display_Info();
System.out.print("ENGINE CAPACITY :"+Engine_Capacity);
}}
public class P26{
public static void main(String[] args){
Scanner sc=new Scanner(System.in);
System.out.print("CHOOSE CAR OR BIKE : ");
String v=sc.nextLine();
```

```
System.out.print("brand :");
String brand=sc.nextLine();
System.out.print("model :");
String model=sc.nextLine();
System.out.print("rental price :");
int price=sc.nextInt();
if (v.equals("Car"))
{
System.out.print("fuel_type :");
sc.nextLine();
String ft=sc.nextLine();
Car car=new Car(brand,model,price,ft);
car.display_Info();
}
else{
System.out.print("engine capacity :");
int et=sc.nextInt();
Bike bike=new Bike(brand,model,price,et);
bike.display_Info();
}
sc.close();
}
}
```

OUTPUT:

```
D:\OOPS\INHERITANCE>java P26
CHOOSE CAR OR BIKE : SWIFT DZIRE
brand :MARUTI SUZUKI
model :X78
rental price :3000
engine capacity :80
BRAND : MARUTI SUZUKI
MODEL : X78
RENTAL PRICE : 3000
ENGINE CAPACITY :80
```

02) A bank wants to develop a Java application to manage different types of bank accounts.
Since all accounts share some common properties, the bank decides to use inheritance to avoid code duplication.
The system should store general account information and also handle specific details for Savings Accounts and Current Accounts.

Base Class: BankAccount
Create a class named BankAccount with the following attributes:
Attributes
- accountNumber – Unique number assigned to the account
- accountHolderName – Name of the account holder
- balance – Current balance in the account

Methods
- deposit(double amount)
  → Adds money to the account balance.
- withdraw(double amount)
  → Withdraws money from the account if sufficient balance is available.
- displayAccountDetails()
  → Displays the account number, account holder name, and balance.

Derived Class: SavingsAccount (inherits from BankAccount)
Create a class SavingsAccount that extends the BankAccount class.
Additional Attribute
- interestRate – Rate of interest given to savings account holders.

Method
- calculateInterest()
  → Calculates interest using the formula:

interest = balance × interestRate / 100
- displaySavingsDetails()
  → Displays account details along with the interest earned.

Derived Class: CurrentAccount (inherits from BankAccount)
Create another class CurrentAccount that extends the BankAccount class.
Additional Attribute
- overdraftLimit – Maximum overdraft limit allowed.

Method
- checkOverdraft()
  → Allows withdrawal even if the balance is insufficient, provided it does not exceed the overdraft limit.
- displayCurrentDetails()
  → Displays account details along with the overdraft limit.

CODE:

```
import java.util.Scanner;
class BankAccount {
int accountNumber;
String accountHolderName;
double balance;
public BankAccount(int accountNumber, String accountHolderName, double
balance) {
this.accountNumber = accountNumber;
this.accountHolderName = accountHolderName;
this.balance = balance;
}
void deposit(double amount) {
balance = balance + amount;
System.out.println("Amount Deposited: " + amount);
}
void withdraw(double amount) {
if (balance >= amount)
balance = balance - amount;
System.out.println("Amount Withdrawn: " + amount);
}
else {
System.out.println("Insufficient Balance");
}}
void displayAccountDetails() {
System.out.println("Account Number: " + accountNumber);
System.out.println("Account Holder Name: " + accountHolderName);
System.out.println("Balance: " + balance);
}}
class SavingsAccount extends BankAccount {
double interestRate;
public SavingsAccount(int accountNumber, String name, double balance,
double interestRate) {
super(accountNumber, name, balance)
this.interestRate = interestRate;
}
void calculateInterest() {
double interest = balance * interestRate / 100;
System.out.println("Interest Earned: " + interest);
}}
class CurrentAccount extends BankAccount {
```

```
double overdraftLimit;
public CurrentAccount(int accountNumber, String name, double balance,
double overdraftLimit) {
super(accountNumber, name, balance);
this.overdraftLimit = overdraftLimit;
}
void checkOverdraft(double amount) {
if (balance + overdraftLimit >= amount) {
balance = balance - amount;
System.out.println("Withdrawal successful using overdraft.");
}
else {
System.out.println("Overdraft limit exceeded.");
}}}
public class P48 {
 public static void main(String[] args) {
Scanner sc = new Scanner(System.in);
System.out.print("Choose Account Type (1-Savings, 2-Current): ");
int choice = sc.nextInt();
System.out.print("Account Number: ");
int accNo = sc.nextInt();
sc.nextLine();
System.out.print("Account Holder Name: ");
String name = sc.nextLine();
System.out.print("Balance: ");
double balance = sc.nextDouble();
if (choice == 1) {
System.out.print("Interest Rate: ");
double rate = sc.nextDouble();

SavingsAccount s = new SavingsAccount(accNo, name, balance, rate);
s.displayAccountDetails();
s.calculateInterest();
}
else if (choice == 2) {
System.out.print("Overdraft Limit: ");
double limit = sc.nextDouble();
CurrentAccount c = new CurrentAccount(accNo, name, balance, limit);
c.displayAccountDetails();
System.out.print("Enter withdrawal amount: ");
double amount = sc.nextDouble();
c.checkOverdraft(amount);
}
```

```
else {
System.out.println("Invalid choice");
}
sc.close();
}}
```

OUTPUT:

```
C:\Users\Nisha\AMRITA\SEM-2\JAVA\Programs>javac P48.java

C:\Users\Nisha\AMRITA\SEM-2\JAVA\Programs>java P48
Choose Account Type (1-Savings, 2-Current): 1
Account Number: 72389427
Account Holder Name: Annie
Balance: 50000
Interest Rate: 5
Account Number: 72389427
Account Holder Name: Annie
Balance: 50000.0
Interest Earned: 2500.0
```